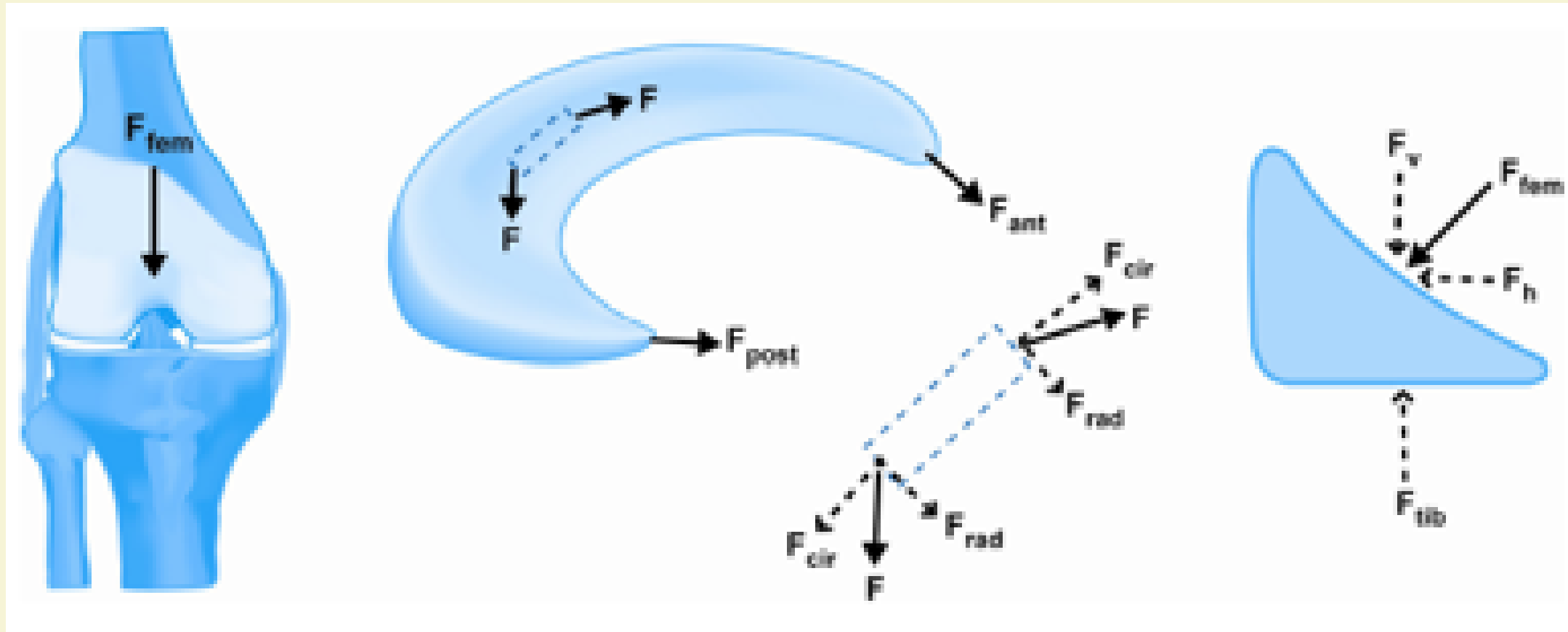


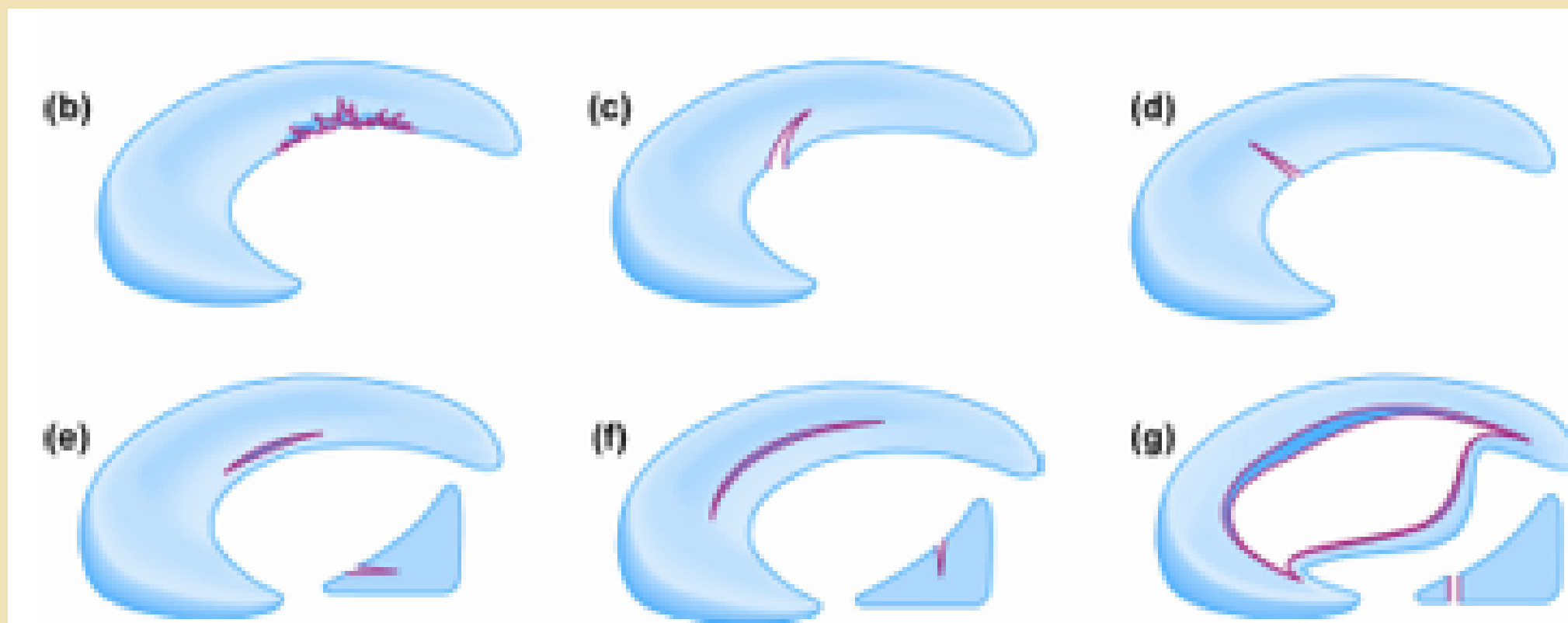
# Mechanical Load Distribution



(Athanasious et. al 2019)

- Axial loads on the femur are transmitted through the meniscus, generating hoop stresses.
- Tensile forces (hoop stresses) resist radial extrusion and help maintain knee joint stability.

## Meniscal Injuries



(Athanasious et. al 2019)

### *Types of Tears*

**Radial Tear:** Small tear from the inner edge outward.

**Longitudinal Tear:** Tear along the meniscus curve.

**Oblique Tear:** Angled tear creating a flap.

**Horizontal Tear:** Split between top and bottom layers.

**Complex Tear:** Combination of multiple tear types.

**Bucket-Handle Tear:** Displaced flap causing joint locking.

